

Kaustubh Hakim

SCIENTIFIC WORK LEADER / RESEARCH STAFF MEMBER

KU Leuven, Institute of Astronomy, Celestijnenlaan 200D, 3001 Leuven, Belgium
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Employment

02.2023 – present **Research Expert (OZK, 0.5 FTE)**, University of Leuven (KU Leuven), Leuven, Belgium
02.2023 – present **Scientific Work Leader (0.5 FTE)**, Royal Observatory of Belgium, Brussels, Belgium
02.2019 – 01.2023 **ERC-funded Postdoctoral Researcher**, University of Bern, Bern, Switzerland
09.2014 – 12.2018 **PhD Candidate**, University of Amsterdam, Amsterdam, Netherlands
07.2014 – 08.2014 **Summer Researcher**, Nicolaus Copernicus Astronomical Center, Warsaw, Poland
07.2010 – 08.2012 **Analyst**, Nomura Investment Bank, Mumbai, India
05.2009 – 07.2009 **Intern**, IBM, Bangalore, India

Career Break

02.2024 – 03.2024 **Paternity Leave**, Belgium

Education

University of Amsterdam / VU Amsterdam

Amsterdam, Netherlands

PHD, ASTROPHYSICS & GEOCHEMISTRY (AWARDED ON 18.12.2018)

12.2018

KU Leuven

Leuven, Belgium

MSc, ASTRONOMY & ASTROPHYSICS

07.2014

Indian Institute of Technology Kharagpur

Kharagpur, India

BTECH, ELECTRONICS ENGINEERING

07.2010

Awards & Distinctions

2025 **FWO Research Project Grant, Belgium**, co-PI (€ 644,480)
2025 **FWO-IRI Grant (MercatObs), Belgium**, co-I (€ 1,659,000)
2023 **BELSP0 FED-TWIN Research Program, Belgium**, Career Award
2021 **NCCR PlanetS, Switzerland**, Research Project Grant
2021 **Swiss Society of Astronomy & Astrophysics**, Travel Award
2020 **University of Bern**, Young Academics Support Award
2014 **KU Leuven**, MSc with *magna cum laude*
2010 **IIT Kharagpur**, BTech with *Honours*
2006 **Engineering Entrance Examination, India**, 99.97 %tile among 0.5 million candidates

Publications

Hakim, K., Bower, D.J., Seidler, F., Sossi, P.A. (2026). Silane–Methane Competition in Sub-Neptune Atmospheres as a Diagnostic of Metallicity and Magma Oceans. *MNRAS* 546, 2 [journal] [arXiv]
Pirvu-Malanda, R., Kaur, S., Viganò, D., et al. (incl. **Hakim, K.**) (2026). Searching for radio emission from stellar wind–magnetosphere interaction or co-rotation breakdown in brown dwarfs. *MNRAS* 547, 3 [journal] [arXiv]
Konings, T., Heinke, L., Baeyens, R., **Hakim, K.**, Christiaens, V., Decin, L. (2025). Reliability of 1D radiative-convective photochemical-equilibrium retrievals on transit spectra of WASP-107b *Astronomy & Astrophysics* 701 A133 [journal] [arXiv]

- Bower, D.J., Thompson, M.A., **Hakim, K.**, Tian, M., Sossi, P.A. (2025). Diversity of Low-mass Planet Atmospheres in the C₂H₂O₂N₂SCl System with Interior Dissolution, Nonideality, and Condensation: Application to TRAPPIST-1e and Sub-Neptunes. *The Astrophysical Journal* 995, 1 [journal] [arXiv]
- Höning, D., Carone, L., Baumeister, P., Chubb, K.L., Grenfell, J.L., **Hakim, K.**, Iro, N., Taysum, B., Tosi, N. (2025). The effect of a biosphere on the habitable timespan of stagnant-lid planets and implications for the atmospheric spectrum. *Astronomy & Astrophysics* 693, A205 [journal] [arXiv]
- Hakim, K.** (2024). Positive Weathering Feedback Compensates Carbonates at Shallow Ocean Depths, IAU Symposium, Vol. 382 (2024) pp. 123–125 [journal]
- Gillmann, C., **Hakim, K.**, Lourenco, D., Quanz, S., Sossi, P. (2024). Interior Controls on the Habitability of Rocky Planets. *Space: Science & Technology* 4, 0075 [journal] [arXiv]
- Narang, M., Oza, A. V., **Hakim, K.**, Puravankara, M., Tyagi, H., Banerjee, B., Surya, A., Nayak, P.K., Banyal, R.K., Thorngren, D.P. (2023), uGMRT observations of the hot-Saturn WASP 69b: Radio-Loud Exoplanet-Exomoon Survey II (RLEES II), *Monthly Notices of the Royal Astronomical Society* 522, 2 [journal] [arXiv]
- Konrad, B.S., Alei, E., Quanz, S.P., Mollière, P., Angerhausen, D., Fortney, J.J., **Hakim, K.**, Jordan, S., Kitzmann, D., Rugheimer, S., Shorttle, O., Wordsworth, R., and the LIFE Collaboration (2023), Large Interferometer For Exoplanets (LIFE): IX. Assessing the impact of clouds on atmospheric retrievals at mid-infrared wavelengths with a Venus-twin exoplanet, *Astronomy & Astrophysics* [journal] [arXiv]
- Hakim, K.**, Tian, M., Bower, D.J., Heng, K. (2023), Diverse Carbonates in Exoplanet Oceans Promote the Carbon Cycle, *Astrophysical Journal Letters* 942, L20 [journal] [arXiv]
- Narang, M., Oza, A. V., **Hakim, K.**, Puravankara, M., Banyal, R., Thorngren, D. (2023), Radio-Loud Exoplanet-Exomoon Survey (RLEES): GMRT Search for Cyclotron Maser Emission, *Astronomical Journal* 165, 1 [journal] [arXiv]
- Bower, D. J., **Hakim, K.**, Sossi, P. A., Sanan, P. (2022), Retention of water in terrestrial magma oceans and carbon-rich early atmospheres, *Planetary Science Journal* 3, 93 [journal] [arXiv]
- Hakim, K.**, Bower, D. J., Tian, M., Deitrick, R., Auclair-Desrotour, P., Kitzmann, D., Dorn, C., Mezger, K., Heng, K. (2021), Lithologic Controls on Silicate Weathering Regimes of Temperate Planets, *Planetary Science Journal* 2, 49 [journal] [arXiv]
- Hakim, K.**, van den Berg, A., Vazan, A., Höning, D., van Westrenen, W., Dominik, C. (2019), Thermal evolution of rocky exoplanets with a graphite outer shell, *Astronomy & Astrophysics* 630, A152 [journal] [arXiv]
- Hakim, K.**, Spaargaren, R., Grewal D.S., Rohrbach A., Brendt J., Dominik, C., van Westrenen, W. (2019), Mineralogy, structure and habitability of carbon-enriched rocky exoplanets: A laboratory approach, *Astrobiology* 19, 7 [journal] [arXiv]
- Hakim, K.**, van Westrenen, W., Dominik, C. (2018), Capturing the oxidation of silicon carbide in rocky exoplanetary interiors, *Astronomy & Astrophysics* 618, L6 [journal] [arXiv]
- Hakim, K.**, Rivoldini, A., Van Hoolst, T., Cottenier, S., Jaeken, J., Chust, T., Steinle-Neumann, G. (2018), A new ab initio equation of state of hcp-Fe and its application to the interior structure and mass-radius relation of rocky super-Earths, *Icarus* 313, 61–78 [journal] [arXiv]

Selected Presentations

- Hakim, K.** (2026, invited seminar), The era of small exoplanets: Need for a multi-disciplinary approach, *Universidad Nacional de Tres de Febrero, Argentina (virtual)*
- Hakim, K.** (2025, invited seminar), Laboratory Investigations of the Interior-Atmosphere Connection of Small Exoplanets, *Observatory of Vienna, Vienna, Austria*
- Hakim, K.** (2025, invited seminar), Chemical Composition of Sub-Neptunes Using a Multi-Disciplinary Approach, *Institute of Science and Technology Austria (ISTA), Klosterneuburg, Austria*
- Hakim, K.** (2025, contributed talk), Influence of Magma on the Atmospheric Chemical Abundances of Hot Sub-Neptunes, *European Planetary Science Congress, Helsinki, Finland*
- Hakim, K.** (2025, contributed talk), The Need for a Multi-Disciplinary Approach for Unravelling the Chemistry of Sub-Neptunes, *Smashing It Workshop, Leeds, United Kingdom*
- Hakim, K.** (2024, invited talk), Exoplanet Ocean Chemistry Under Carbonate-Silicate Cycling, *Venus Workshop, Austrian Academy of Sciences, Graz, Austria*

Hakim, K. (2024, contributed talk), Magma-Atmosphere Coupling of Hot Sub-Neptunes, *Ariel Consortium Meeting, Lisbon, Portugal*

Hakim, K. (2024, contributed talk), Magma-Atmosphere Coupling of Sub-Neptune Exoplanets, *Geologica Belgica Luxembourg Meeting, Liege, Belgium*

Hakim, K. (2024, invited seminar), Magma-Atmosphere Coupling in Hot Sub-Neptunes, *Institute for Particle Physics and Astrophysics, ETH Zurich, Switzerland*

Hakim, K. (2024, contributed talk), Impact of Magma-Atmosphere Coupling on Sub-Neptune Populations, *COST Action PLANETS Kick-Off, Torun, Poland*

Hakim, K. (2024, contributed poster), Sub-Neptune populations from magma-atmosphere coupling, *Exoplanets-5 Conference 2024, Leiden, Netherlands*

Hakim, K. (2024, invited talk), The Need for Ab Initio and Laboratory Simulations in Exoplanet Science, *Netherlands Astronomers' Conference 2024, Egmond aan Zee, Netherlands*

Hakim, K. (2024, invited lecture), Chemical Diversity in Rocky Exoplanets & Sub-Neptunes from Laboratory and Ab Initio Simulations 2024 ATSOA Summer Training School, *ARIES, Nainital, India*

Hakim, K. (2024, invited seminar), Chemical Diversity in Rocky Interiors and Interior-Atmosphere Interactions, *Exoplanet Seminar Series, NASA Goddard Space Flight Center, USA (virtual)*

Hakim, K. (2023, invited lecture), Chemistry of Rocky Worlds and Sub-Neptunes, *ATOMIUM Meeting, KU Leuven, Belgium*

Hakim, K. (2023, invited review talk), Astronomical Windows into Planetary Interiors and Evolution, *Strange New Worlds Exoplanet Conference, IISER Pune, India*

Hakim, K. (2023, invited talk), Impact of Refractory Carbon on Rocky Exoplanet Mineralogy, Evolution & Habitability, *VLT MATISSE Science Team Meeting (virtual)*

Hakim, K. (2022, invited talk), Chemically-Diverse Exoplanet Interiors and Interior-Atmosphere Interactions, *NOVA NW2 Meeting, Groningen, Netherlands (virtual)*

Hakim, K. (2022, invited talk), Carbon and Sulfur in Exoplanetary Interiors, *HP4 Workshop, Brussels, Belgium*

Hakim, K. (2022, invited review talk), Exoplanetary Interiors and Evolution, *PFE-SPP1992 joint meeting, Berlin, Germany*

Hakim, K. (2022, invited seminar), Chemistry of Planetary Interiors and Surfaces, *National Institute of Science Education and Research, Bhubaneswar, India*

Hakim, K. (2020, plenary talk), Geochemistry of Carbon Cycles on Rocky Exoplanets, *Exoplanets III Conference (virtual)*

Hakim, K. (2020, invited seminar), Application of Geosciences to Exoplanets, *Tata Institute of Fundamental Research, Mumbai, India*

Hakim, K. (2019, poster), Geochemistry of Carbon Cycles on Rocky Exoplanets, *Extreme Solar Systems 4, Reykjavik, Iceland*

Hakim, K., van den Berg, A., Vazan, A., Höning, D., van Westrenen, W., Dominik, C. (2019, contributed talk), Thermal evolution of rocky exoplanets covered with graphite, *Division of Planetary Sciences – Europlanet Science Congress, Geneva, Switzerland*

Hakim, K. (2017, seminar), A laboratory approach to probe the mineralogy of carbon-enriched rocky exoplanets, *University of Chicago, USA*

Hakim, K., van Westrenen, W., Dominik, C. (2017, contributed talk), Mineralogy of Carbon-Enriched Rocky Extra-Solar Planets from Laboratory Experiments, *Lunar and Planetary Science Conference, Houston, Texas*

Hakim, K., Rivoldini, A., Van Hoolst, T., Cottenier, S., Chust, T., Steinle-Neumann, G. (2017, contributed talk), A New Ab Initio Equation of State of hcp-Iron and Its Application to the Interior Structure of Rocky Super-Earths, *Lunar and Planetary Science Conference, Houston, Texas*

Teaching Experience

2024 – present	Physics and Chemistry of Planets , Co-Lecturer, <i>MSc Astro. & Astrophy.</i> , KU Leuven
2020 – 2021	Radiative Transfer , Teaching Assistant / Co-Lecturer, <i>MSc Physics</i> , University of Bern
2018	Planetary Science , Co-Lecturer, <i>MSc Earth Sciences</i> , VU Amsterdam
2015 – 2017	Interstellar Medium , Teaching Assistant, <i>MSc Astro. & Astrophy.</i> , University of Amsterdam

Supervision Experience

Currently supervising 3 PhD and 3 MSc students; co-supervising/collaborating with 3 further PhD students. Previously supervised 8 MSc and 6 BSc students.

2026 – present	Anjana Shaju , PhD Thesis, KU Leuven (Daily Supervisor)
2025 – present	Luka Vranckx , PhD Thesis, Royal Observatory of Belgium / KU Leuven (Daily Supervisor)
2025 – present	Bohao Chen , PhD Thesis, KU Leuven (Daily Supervisor)
2026 – present	Janne Moons , MSc Thesis, KU Leuven (Supervisor)
2024 – 2025	Darshan Gove , MSc Thesis, KU Leuven (Supervisor)
2024 – present	Soetkin Willemyns , PhD Thesis, KU Leuven (Co-Supervisor)
2023 – present	Nuno Pereira , PhD Thesis, KU Leuven (Co-Supervisor)
2023 – 2024	Ine Malfait , MSc Thesis, KU Leuven (Supervisor)
2021 – 2022	Lukas Carmichael , MSc Thesis, ETH Zürich (Co-Supervisor)
2018	Dieke Bentjees , BSc Thesis, VU Amsterdam (Co-Supervisor)
2016	Rob Spaargaren , BSc Thesis, VU Amsterdam (Co-Supervisor)

Service, Leadership and Professional Development

REVIEWING SERVICE FOR FUNDING AGENCIES

NASA ROSES

Université Paris Cité

Swiss National Science Foundation

Israel Science Foundation (ISF)

REVIEWING SERVICE FOR HIGH-IMPACT JOURNALS

Astrobiology

Journal of Geophysical Research

Astronomy & Astrophysics

Monthly Notices of Royal Astronomical Society

Planetary Science Journal

Geochimica et Cosmochimica Acta

Open Research Europe

Applied Geochemistry

Frontiers in Earth Science

Nature

Reviews of Modern Physics

LEADERSHIP ROLES

2026	Local Organising Committee (LOC) , Mercury Conference, Leuven, Belgium
2025	Scientific Organising Committee (SOC) , Belgian-Indian Astronomy Meeting, Calicut, India
2023	Session Convener , Goldschmidt Conference, Lyon, France
2022	Working Group Coordinator , NCCR PlanetS Phase 3
2022 – 2023	Session Convener , European Geosciences Union General Assembly, Vienna, Austria
2019 – 2022	Working Group Organiser , Atmosphere-Interior Exchange, University of Bern
2016 – 2017	PhD Representative , PhD-PostDoc Council, University of Amsterdam, Netherlands
2015 – 2017	Meeting Organiser , Interdisciplinary PEPSci Network, Netherlands
2012 – 2014	Student Representative , Permanent Education Committee, KU Leuven

SELECTED OUTREACH AND EDUCATION

2024 **Co-Lecturer**, Starquakes and Exoplanets in the Milky Way, Junior College STEM Course, KU Leuven

2023 **Podcast Guest**, Dark Matter Podcast on YouTube, Friends of Europe, Brussels

- 2021 **Outreach Contribution**, Planets in the Solar System, Astronomy Calendar
- 2019 **Public Science Talk**, Cooking Tiny Planets in the Lab, Astronomy on Tap Bern
- 2019 **Blog Writing**, The Role of Geosciences in Exoplanet Science, European Geosciences Union Blogs
- 2018 **Magazine Contribution**, A New Ultra-High-Pressure Equation Of State For Iron, Science Trends
- 2016 **Student Engagement Talk**, Carbon-rich exoplanets, Universidad de los Andes, Bogotá, Colombia
- 2015 **Volunteer**, Public Stargazing Nights, University of Amsterdam

PROFESSIONAL MEMBERSHIPS

- 2025 – present **Member, Science Team**, METIS instrument, Extremely Large Telescope
- 2018 – present **Member**, International Astronomical Union
- 2019 – present **Member**, European Geosciences Union
- 2020 – present **Member**, European Association of Geochemistry